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These solutions are not the only "perfect" or "10/10" solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Q1. Consider a plane $\mathcal{P} = \{\mathbf{x} : \langle \mathbf{n}, \mathbf{x} \rangle = 0\}$ defined by a non-zero normal vector $\mathbf{n} \neq \mathbf{0}$.

(a) Give a precise definition of the linear transformation:

$$\sigma_{\mathcal{P}} = \text{“reflection across the plane } \mathcal{P}\text{”}$$

Solution: Let $\mathbf{v}$ and $\mathbf{w}$ be two independent vectors in $\mathcal{P}$ such that $\langle \mathbf{v}, \mathbf{w} \rangle = 0$, and let's assume that $\|\mathbf{v}\| = \|\mathbf{w}\| = 1$. Consider now the linear transformation $T \colon \mathbb{R}^3 \to \mathbb{R}^3$ given by

$$T(\mathbf{u}) := \langle \mathbf{u}, \mathbf{v} \rangle \mathbf{v} + \langle \mathbf{u}, \mathbf{w} \rangle \mathbf{w} - \langle \mathbf{u}, \mathbf{n} \rangle \mathbf{n}.$$

Note that:

1. $T(\mathbf{v}) = \langle \mathbf{v}, \mathbf{v} \rangle \mathbf{v} + \langle \mathbf{v}, \mathbf{w} \rangle \mathbf{w} - \langle \mathbf{v}, \mathbf{n} \rangle \mathbf{n} = \mathbf{v}$.
2. $T(\mathbf{w}) = \langle \mathbf{w}, \mathbf{v} \rangle \mathbf{v} + \langle \mathbf{w}, \mathbf{w} \rangle \mathbf{w} - \langle \mathbf{w}, \mathbf{n} \rangle \mathbf{n} = \mathbf{w}$.
3. $T(\mathbf{n}) = \langle \mathbf{n}, \mathbf{v} \rangle \mathbf{v} + \langle \mathbf{n}, \mathbf{w} \rangle \mathbf{w} - \langle \mathbf{n}, \mathbf{n} \rangle \mathbf{n} = -\mathbf{n}$.

Since the linear transformations T and $\sigma_{\mathcal{P}}$ coincide in the basis $\{\mathbf{v}, \mathbf{w}, \mathbf{n}\}$ for $\mathbb{R}^3$, they coincide in the whole space $\mathbb{R}^3$.

(b) Prove that $\sigma_{\mathcal{P}} \circ \sigma_{\mathcal{P}}$ is the identity transformation.

Solution: Note that

1. $\sigma_{\mathcal{P}}(\sigma_{\mathcal{P}}(\mathbf{v})) = \sigma_{\mathcal{P}}(\mathbf{v}) = \mathbf{v}$.
2. $\sigma_{\mathcal{P}}(\sigma_{\mathcal{P}}(\mathbf{w})) = \sigma_{\mathcal{P}}(\mathbf{w}) = \mathbf{w}$.
3. $\sigma_{\mathcal{P}}(\sigma_{\mathcal{P}}(\mathbf{n})) = \sigma_{\mathcal{P}}(-\mathbf{n}) = -\sigma_{\mathcal{P}}(\mathbf{n}) = \mathbf{n}$.

Since the identity and the linear transformation $\sigma_{\mathcal{P}} \circ \sigma_{\mathcal{P}}$ coincide in the basis $\{\mathbf{v}, \mathbf{w}, \mathbf{n}\}$ for $\mathbb{R}^3$, they coincide in the whole space $\mathbb{R}^3$.

Q2. Consider a basis $\alpha = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V. Write matrices $[T]_\alpha^\alpha$ for the following linear transformations

(a) For some $1 \leq k < n$ we define:

$$T(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k + \cdots + a_n\mathbf{v}_n) = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k$$

Solution: Note that if $1 \leq k < n$ is fixed, then

$$T(\mathbf{v}_i) = \begin{cases} \mathbf{v}_i & \text{if } i \leq k \\ \mathbf{0} & \text{otherwise} \end{cases}$$

Therefore, the matrix representation of T with respect to α is

$$[T]_\alpha^\alpha = \begin{pmatrix} 1 & 0 & \dots & \dots & 0 \\ 0 & 1 & \dots & \dots & 0 \\ \vdots & \vdots & & & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & & & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix}$$

where only k-many rows include a 1 in the diagonal entry.

(b)

$$T(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n) = a_n\mathbf{v}_1 + a_{n-1}\mathbf{v}_2 + \cdots + a_1\mathbf{v}_n$$

Solution: Note that each $\mathbf{v}_i = \delta_{i1}\mathbf{v}_1 + \delta_{i2}\mathbf{v}_2 + \cdots + \delta_{in}\mathbf{v}_n$ where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Thus, $T(\mathbf{v}_i) = \delta_{in}\mathbf{v}_1 + \delta_{i(n-1)}\mathbf{v}_2 + \cdots + \delta_{i2}\mathbf{v}_{n-1} + \delta_{i1}\mathbf{v}_n = \mathbf{v}_{n-i+1}$, and therefore

$$[T]_\alpha^\alpha = \begin{pmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 1 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \end{pmatrix}$$

Q3. Give a specific example of a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ and bases $\alpha = \{\mathbf{v}_1, \mathbf{v}_2\}$ and $\beta = \{\mathbf{w}_1, \mathbf{w}_2\}$ such that $[T]_\alpha^\alpha \neq [T]_\beta^\beta$.

Solution: Let $P\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the projection onto the x-axis, and let

$$\alpha = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \beta = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

Then

$$[P]_\alpha^\alpha = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad [P]_\beta^\beta = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Moral: The order matters!

Q4. Consider two finite-dimensional (real) vector spaces V and W with $\dim(V) = n$ and $\dim(W) = k$. Define $\mathcal{L}(V, W)$ to be the set of linear transformations:

$$\mathcal{L}(V, W) = \{T : \text{ The map } T : V \to W \text{ is linear } \}.$$

(a) Define addition and scaling on $\mathcal{L}(V, W)$.

Solution: As with functions, we can define the addition and scaling as:

$$(T + S)(\mathbf{v}) := T(\mathbf{v}) + S(\mathbf{v}) \quad \text{and} \quad (cT)(\mathbf{v}) := c(T(\mathbf{v})).$$

Note that these operations are well-defined, since:

1. If both T and S are elements of $\mathcal{L}(V, W)$, then

$$\begin{aligned}(T + S)(\mathbf{u} + k\mathbf{v}) &= T(\mathbf{u} + k\mathbf{v}) + S(\mathbf{u} + k\mathbf{v}) \\ &= (T(\mathbf{u}) + S(\mathbf{u})) + k(T(\mathbf{v}) + S(\mathbf{v})) \\ &= (T + S)(\mathbf{u}) + k(T + S)(\mathbf{v})\end{aligned}$$

for all $\mathbf{u}, \mathbf{v} \in V$ and every $k \in \mathbb{R}$, so $T + S$ is in $\mathcal{L}(V, W)$ as well.

2. Also, if $T \in \mathcal{L}(V, W)$ and $c \in \mathbb{R}$, then

$$(cT)(\mathbf{u} + k\mathbf{v}) = c(T(\mathbf{u} + k\mathbf{v})) = cT(\mathbf{u}) + ckT(\mathbf{v}) = (cT)(\mathbf{u}) + k(cT)(\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$ and every $k \in \mathbb{R}$, so $cT \in \mathcal{L}(V, W)$.

(b) Sketch a proof that $\mathcal{L}(V, W)$ is a vector space.

Solution: Given how the operations are defined, the argument is pretty much the same as proving that $F(\mathbb{R})$ is a vector space.

(c) Prove that $\mathcal{L}(V, W)$ is finite-dimensional by finding an explicit basis for it.

Solution: Fix bases $\alpha = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\beta = \{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ for V and W respectively, and consider, for each $i \leq n$ and $j \leq k$, the linear transformation determined by

$$T_{ij}(\mathbf{v}_i) = \mathbf{w}_j \text{ and } T_{ij}(\mathbf{v}_k) = \mathbf{0} \text{ if } i \neq k.$$

Claim 1: The set $\{T_{ij} : i \leq n \text{ and } j \leq k\}$ is linearly independent.
Proof: If we have a null linear combination

$$\sum_{i \leq n,\, j \leq k} a_{ij} T_{ij} = Z \quad \text{(here } Z \text{ denotes the zero linear transformation)},$$

then we can evaluate the equation above at each vector of the basis α, and obtain that

$$\mathbf{0}_W = Z(\mathbf{v}_i) = \sum_{j \leq k} a_{ij} T_{ij}(\mathbf{v}_i) = a_{i1}\mathbf{w}_1 + \cdots + a_{ik}\mathbf{w}_k.$$

Since β is linearly independent, we can conclude that all the a_{ij}'s equal 0.

Claim 2: The set $\{T_{ij} : i \leq n \text{ and } j \leq k\}$ spans $\mathcal{L}(V, W)$.

Proof: Let $T\colon V \to W$ be linear, and for each $i \leq n$ let $a_{i1}, \dots, a_{ik} \in \mathbb{R}$ be such that

$$T(\mathbf{v}_i) = a_{i1}\mathbf{w}_1 + \cdots + a_{ik}\mathbf{w}_k = a_{i1}T_{i1}(\mathbf{v}_i) + \cdots + a_{ik}T_{ik}(\mathbf{v}_i) = (a_{i1}T_{i1} + \cdots + a_{ik}T_{ik})(\mathbf{v}_i).$$

Since the linear transformations T and

$$\sum_{i \leq n,\, j \leq k} a_{ij}T_{ij}$$

coincide in α, then they are equal in the whole space V.

We can conclude that the set $\{T_{ij} : i \leq n \text{ and } j \leq k\}$ is a basis for $\mathcal{L}(V, W)$.

Q5. Consider $M_{n\times n}(\mathbb{R})$, the set of $n \times n$ matrices with real entries, and define

$$T\colon M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R})$$

by $T([m_{ij}]) = [m_{ji}]$.

(a) Prove that T is linear.

Solution: Note that given matrices $[m_{ij}]$ and $[n_{ij}]$ in $M_{n\times n}(\mathbb{R})$, and $k \in \mathbb{R}$, we have that

$$T([m_{ij}] + k[n_{ij}]) = T([m_{ij} + kn_{ij}]]) = [m_{ji} + kn_{ji}] = [m_{ji}] + k[n_{ji}] = T([m_{ij}]) + kT([n_{ij}]).$$

(b) Prove that

$$\mathrm{Symm}_{n\times n}(\mathbb{R}) = \{M \in M_{n\times n}(\mathbb{R}) : T(M) = M\} \text{ and } \mathrm{Skew}_{n\times n}(\mathbb{R}) = \{M \in M_{n\times n}(\mathbb{R}) : T(M) = -M\}$$

are vector subspaces of $M_{n\times n}(\mathbb{R})$.

Solution: We will do the proof for $\mathrm{Symm}_{n\times n}(\mathbb{R})$. The proof for $\mathrm{Skew}_{n\times n}(\mathbb{R})$ is analogous:

1. Since T is linear, we have that $T(\mathbf{0}) = \mathbf{0}$, so $\mathbf{0} \in \mathrm{Symm}_{n\times n}(\mathbb{R})$.
2. If $M, N \in \mathrm{Symm}_{n\times n}(\mathbb{R})$, and $k \in \mathbb{R}$, then $T(M + kN) = T(M) + kT(N) = M + kN$, so $M + kN \in \mathrm{Symm}_{n\times n}(\mathbb{R})$ as well.

Reflection: Note that the very same technique applied in this proof also works to prove a way more general result: *If $T, S\colon V \to W$ are linear transformations, then the set*

$$U := \{\mathbf{v} \in V : T(\mathbf{v}) = S(\mathbf{v})\}$$

is a vector subspace of V.
With what you learned this week + Q4 above, can you express U as a kernel?

(c) Calculate the dimensions of $\mathrm{Symm}_{n\times n}(\mathbb{R})$ and $\mathrm{Skew}_{n\times n}(\mathbb{R})$.

Solution: We will do the computation for $\mathrm{Symm}_{n\times n}(\mathbb{R})$. The one for $\mathrm{Skew}_{n\times n}(\mathbb{R})$ is analogous: If a matrix M is symmetric, then the ij-th entry of M has to be equal to its ji-ith entry for every $i, j \leq n$. So, symmetric matrices are of the form

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{12} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{13} & a_{23} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \dots & a_{nn} \end{pmatrix}$$

This implies that we have n-many degrees of freedom for the diagonal entries, but only $(n^2-n)/2$ (the upper half of the matrix) degrees of freedom outside of the diagonal. In total, we have

$$\frac{n^2-n}{2} + n = \frac{n^2+n}{2}$$

entries to choose our coefficients, and therefore $\dim(\mathrm{Symm}_{n\times n}(\mathbb{R})) = (n^2+n)/2$.

Q6. Suppose that $T : V \to W$ is a linear map. Prove that T is one-to-one if, and only if, T maps linearly independent sets in V to linearly independent sets in W.

Solution: ($\Rightarrow$) Let $T\colon V \to W$ be one-to-one, and let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a linearly independent subset of V. To prove that the set $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}$ is also linearly independent, consider an arbitrary null linear combination

$$a_1T(\mathbf{v}_1) + \cdots + a_nT(\mathbf{v}_n) = \mathbf{0}.$$

Since T is linear, we have that

$$T(a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n) = \mathbf{0}.$$

Finally, since $T(\mathbf{0}) = \mathbf{0}$, and T is one-to-one, we can conclude that

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}.$$

The linear independence of the set of $\mathbf{v}_i$'s implies that $a_1 = \cdots = a_n = 0$.

($\Leftarrow$) Let $\mathbf{v}$ and $\mathbf{w}$ be different elements of V so that $\mathbf{v} - \mathbf{w} \neq \mathbf{0}$. In particular, the singleton $\{\mathbf{v} - \mathbf{w}\}$ is linearly independent, and therefore $\{T(\mathbf{v} - \mathbf{w})\}$ is linearly independent as well. This, of course, implies that $T(\mathbf{v} - \mathbf{w}) \neq \mathbf{0}$, or equivalently, that $T(\mathbf{v}) \neq T(\mathbf{w})$. Thus, T is one-to-one.

Q7. Consider

$$\mathbb{R}^\omega = \{(x_0, x_1, \dots) : x_i \in \mathbb{R}\}$$

the vector space of countably infinite sequences of reals. We define the following maps:

$$L(x_0, x_1, x_2, \dots) = (x_1, x_2, \dots) \qquad R(x_0, x_1, x_2, \dots) = (0, x_0, x_1, x_2, \dots)$$

we call L the <u>left shift</u> transformation and we call R the <u>right shift</u> transformation.

(a) Prove that L is onto but not one-to-one.

Solution: Note that for every sequence $(x_0, \dots, x_n, \dots)$, we have that

$$(x_0, \dots, x_n, \dots) = L(0, x_0, \dots, x_n, \dots),$$

so L is onto. Nevertheless,

$$L(0, 0, 0, \dots, 0, \dots) = L(1, 0, 0, \dots, 0, \dots),$$

so L is not one-to-one.

(b) Prove that R is one-to-one but not onto.

Solution: If two sequences $(x_0, \dots, x_n, \dots)$ and $(y_0, \dots, y_n, \dots)$ are different, there is some $i \in \mathbb{N}$ such that $x_i \neq y_i$. But this still is true for the sequences

$$R(x_0, \dots, x_n, \dots) = (0, x_0, \dots, x_n, \dots) \text{ and } R(y_0, \dots, y_n, \dots) = (0, y_0, \dots, y_n, \dots),$$

so R is one-to-one. Also, every sequence in the image of R starts with a 0, so R cannot be onto.

(c) Why are these transformations special? Could such maps exist for a finite-dimensional space?

Solution: Maps like these (onto but not one-to-one, or one-to-one but not onto) cannot exist in the finite-dimensional framework. This is a consequence of the Dimension Theorem that we will cover later in the course.

Q8. Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis of $\mathbb{R}^3$. Let $T : \mathbb{R}^3 \to \mathbb{R}$ be a linear transformation. Show that there are constants $a, b, c \in \mathbb{R}$ such that:

$$T(x\mathbf{v}_1 + y\mathbf{v}_2 + z\mathbf{v}_3) = ax + by + cz$$

Solution: Consider $a = T(\mathbf{v}_1)$, $b = T(\mathbf{v}_2)$ and $c = T(\mathbf{v}_3)$. Then

$$T(x\mathbf{v}_1 + y\mathbf{v}_2 + z\mathbf{v}_3) = xT(\mathbf{v}_1) + yT(\mathbf{v}_2) + zT(\mathbf{v}_3) = ax + by + cz.$$